

Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_no_replay.
- Best CVRPLIB holdout gap: cvrp_no_replay.
- Best synthetic holdout gap: cvrp_failure_replay.

Run Metadata

- run_name: run_20260513_193644_l
- started_at_local: 2026-05-13 19:36:44
- finished_at_local: 2026-05-13 19:39:46
- duration_hhmm: 00:03
- duration_seconds: 182.572
- seed_offset: 11000
- replicate_label: l
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.225234	0.01914	0.133637	0.783611	0.78	0.010307
cvrp_random_replay	random	score_only	False	0.227843	0.01914	0.135086	0.875321	0.78	0.019345
cvrp_failure_replay	failure	score_only	False	0.245738	0.005376	0.13891	0.863841	0.78	0.015991
cvrp_failure_replay_compression	failure	novelty_gate	True	0.256804	0.005376	0.145058	0.887047	0.58	0.012532

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.225234, synthetic 0.01914, combined 0.133637.

- Accepted-epoch count 2, mean accepted code novelty 0.783611, and final complexity 0.78.
- Adaptation efficiency 0.010307 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.225234 across 5 instances; family means: A=0.091623, B=0.244087, E=0.293666, P=0.252708.
- Panel synthetic_holdout mean gap 0.01914 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.055054.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 562, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.227074}, {"best_known_cost": 672, "cost": 808, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.202381}, {"best_known_cost": 784, "cost": 863, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.100765}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.227843, synthetic 0.01914, combined 0.135086.
- Accepted-epoch count 2, mean accepted code novelty 0.875321, and final complexity 0.78.
- Adaptation efficiency 0.019345 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.227843 across 5 instances; family means: A=0.109075, B=0.244761, E=0.287908, P=0.252708.
- Panel synthetic_holdout mean gap 0.01914 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.055054.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 609, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.329694}, {"best_known_cost": 937, "cost": 1187, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.266809}, {"best_known_cost": 672, "cost": 808, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.202381}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.245738, synthetic 0.005376, combined 0.13891.
- Accepted-epoch count 2, mean accepted code novelty 0.863841, and final complexity 0.78.
- Adaptation efficiency 0.015991 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.245738 across 5 instances; family means: A=0.086387, B=0.27715, E=0.295585, P=0.292419.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 611, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.334061}, {"best_known_cost": 937, "cost": 1236, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.319104}, {"best_known_cost": 672, "cost": 817, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.215774}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.

- Final transfer gaps: CVRPLIB 0.256804, synthetic 0.005376, combined 0.145058.
- Accepted-epoch count 4, mean accepted code novelty 0.887047, and final complexity 0.58.
- Adaptation efficiency 0.012532 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.256804 across 5 instances; family means: A=0.115183, B=0.259605, E=0.330134, P=0.319495.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1057, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.128068}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

Judge Appendix

Summary of Results by Condition

Condition	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Complexity	Replay Mode	Selection Mode	Code Novelty (mean)	Archive Size	Compression Pressure	Adaptation Efficiency
cvrp_no_replay	**0.225234**	0.01914	**0.133637**	0.78	none	score_only	0.7836	14	no	0.01031
cvrp_random_replay	0.227843	0.01914	0.135086	0.78	random	score_only	0.8753	14	no	0.01935
cvrp_failure_replay	0.245738	**0.005376**	0.13891	0.78	failure	score_only	0.8638	14	no	0.01599
cvrp_failure_replay_compression	0.256804	**0.005376**	0.145058	**0.58**	failure	novelty_gate	0.8870	14	yes	0.01253

Interpretation

****Main Transfer Evidence:****

- The best final CVRPLIB held-out gap and transfer gap are achieved by the ****no replay**** condition (0.225234 heldout gap; 0.133637 transfer gap), indicating superior generalization to CVRPLIB instances and transfer performance.
- Synthetic holdout gap is lowest in both ****failure replay**** conditions (0.005376), indicating they provide better synthetic domain generalization but not better CVRPLIB transfer.

****Replay Modes:****

- ****No replay**** yields the best transfer performance despite having the lowest code novelty (0.7836).
- ****Random replay**** increases code novelty (0.8753) and adaptation efficiency (0.01935), but with slightly worse CVRPLIB transfer metrics compared to no replay.
- ****Failure replay**** (both score_only and novelty_gate selection) shows increased code novelty (~0.86-0.88), better synthetic generalization, but degrades CVRPLIB transfer performance.

- **Compression-aware failure replay** reduces complexity significantly (0.58 vs 0.78) with no improvement in transfer or heldout gaps, indicating compression reduces code complexity but not transfer performance.

Algorithmic Novelty vs Lexical Novelty:

- Code novelty is higher with replay approaches but does **not** coincide with improved transfer to CVRPLIB, suggesting lexical novelty increases without corresponding transfer gains.
- The best transfer performance (no replay) occurs with **lower code novelty**, supporting the conservative interpretation that no replay yields more robust transfer despite less exploratory diversity.

Summary:

- **No replay condition provides best heldout CVRPLIB gap and overall transfer gap, representing strongest transfer performance.**
- **Failure replay strategies improve synthetic domain generalization but at the cost of worse CVRPLIB transfer gaps.**
- **Increased replay modes lead to higher code novelty but do not translate into improved transfer performance.**
- **Compression-aware replay reduces code complexity but further degrades CVRPLIB transfer.**

Recommendations

- Prioritize **no replay** setting for benchmarks focused on CVRPLIB transfer and heldout generalization.
- Consider failure replay only if synthetic holdout performance is critical, acknowledging it harms CVRPLIB transfer.
- Lexical/code novelty is not evidence of algorithmic improvement unless coupled with better transfer metrics; here it is not.
- Compression affects complexity but not transfer; use cautiously when aiming for generalization in CVRP benchmarks.